

The Tolerance Law:

Contact-Rich Assembly Skills Have a Capacity-Dependent Learnability Phase Transition

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Summary. A robot either can or cannot learn to insert a part, and the boundary depends on an engineering parameter the factory already controls: the *clearance* between peg and channel. We vary clearance across four levels and measure when behavior cloning succeeds, using a controlled contact-rich insertion task in MuJoCo with realistic physics. Below a task-specific clearance c^* , the learned policy jams with near-certainty; above it, insertion is reliable. The boundary is *not* the teacher’s failure—our expert succeeds on 94.8% of episodes at the tightest clearance. The second and more surprising result is that the boundary depends on model capacity in a *non-monotone* way: increasing the width of the behavior-cloning network from $w=32$ to $w=128$ pushes the boundary downward (more clearances become learnable), but increasing further to $w=256$ pushes it *back up*—the larger network overfits demonstration noise and degrades on held-out episodes. We term this the **Tolerance Law**: the learnability of contact-rich assembly skills is a phase transition in an engineering parameter, and the transition boundary is a non-monotone function of model capacity, with a measurable optimal operating point.

1 Introduction

Assembly is the least automated step in manufacturing. The same task that a human can do by feel remains beyond the reach of scripted robots when part variation, fixture drift, and measurement noise collide at a chamfer. Learning from demonstration (LfD) is the natural answer: a robot watches an expert insert a part and imitates. Yet industrial deployment of LfD remains rare—and the reason usually given is data: there is never enough of it.

We present evidence for a different and more useful explanation. In a controlled contact-rich insertion task with real MuJoCo physics, we vary *one* engineering parameter—the lateral clearance c between a peg and its channel—and measure when behavior cloning succeeds. The result is not a gradual degradation but a *boundary*: below a clearance c^* , the learned policy jams with near-certainty; above it, insertion is reliable. This boundary exists even when the expert teacher succeeds on every clearance, which isolates the learner as the cause.

The second result is that the boundary depends on model capacity in a *non-monotone* way. At fixed data budget, increasing the width of the behavior-cloning MLP from $w=32$ to $w=128$ pushes the learnability boundary downward. But increasing further to $w=256$ pushes it *back up*: the larger network overfits the oscillatory demonstration noise and degrades on held-out episodes. The phase diagram shows a crossing—at tight clearance, a mid-capacity network outperforms both smaller and larger ones. An adaptive capacity controller that walks width upward and stops when learning succeeds correctly identifies $w=128$ as the minimum effective capacity at each clearance—a capacity *sweet spot* the factory can find online.

We call the aggregate the *Tolerance Law*: in assembly LfD, learnability is a phase transition in an engineering parameter, the transition boundary depends on model capacity, and the relationship is non-monotone—bigger is not always better. If it holds beyond this planar task, it reframes the capacity-planning problem in manufacturing robotics: the question is not only *how much data* but *what capacity that data buys*, and the answer has a sweet spot the factory can discover.

2 Task and Teacher

Environment. A two-axis gantry stage carries a rigid peg (50×50 mm face) toward a slot whose channel width is $2(\frac{1}{2}w_{\text{peg}} + c)$, so c is the per-side clearance in metres. The slot sits on a fixture that is rigidly locked during each episode; its true position y_{ch} is sampled uniformly in ± 0.10 m and the robot observes a noisy measurement $y_{\text{meas}} \sim \mathcal{N}(y_{\text{ch}}, \sigma_y^2)$ with $\sigma_y = 0.8$ mm. Observations are $[\text{peg}_x, \text{peg}_y, \dot{x}, \dot{y}, y_{\text{meas}}, F_x, F_y, t]$ (8-dimensional) and actions are velocity commands integrated by position actuators (150 N/m axial, 400 N/m lateral, 80 N cap). Contact is resolved by MuJoCo’s constraint solver at 200 Hz; the peg can jam, rub, and ratchet against the mouth edges. Four clearances span $c = 0.5$ to 4.0 mm.

The teacher is force-blind. The expert aligns to the measured slot centre, then inserts while sweeping laterally with a deterministic time-based sinusoid (amplitude matched to the sensor noise) and, on a stall, retracting and re-approaching with a shifted sweep bias (up to 16 attempts). It never reads the contact forces. Its success is 94.8% at the tightest clearance and 100% at the loosest—measured over 10 independent seeds and 40 evaluation episodes each. The teacher thus generates learnable demonstrations at every clearance, so any failure below c^* is a failure of the learner, not of the data. The sweep is a deterministic function of time (which is in the observation), so a behavior-cloned policy can in principle reproduce it; its *fitted amplitude* is the quantity that is fragile to capacity mismatches.

Learners. Behavior cloning by regression (MSE on actions) with an MLP whose width $w \in \{32, 128, 256\}$ and depth $d = 3$ are the capacity axis (3 width choices). Policies are trained on N successful demonstrations ($N \in \{20, 60, 160\}$, 3 budget levels) and evaluated on $n_{\text{eval}} = 40$ fresh episodes with 10 seeds (360 cells in total). Success is a fully seated peg held for 10 steps. We report mean $\pm$ standard error across seeds.

3 Results

3.1 The capacity–data phase diagram

Figure 1 shows learned success as a function of clearance, capacity, and budget. Three facts stand out.

First, there is a learnability boundary: at any fixed (capacity, budget), success drops steeply across a narrow clearance band. The boundary is not an artifact of a weak teacher—the teacher succeeds at every clearance (Fig. 1, top row).

Second, the boundary is non-monotone in width. At $N=60$ (medium budget), $w=128$ outperforms both $w=32$ and $w=256$ at tight clearance. At $N=160$ (large budget), the same ordering holds—and $w=256$ *worsens* relative to its smaller self at the tightest clearance, confirming overfitting.

Third, the data budget shifts the boundary but with capacity-dependent diminishing returns. At $w=128$, all budgets reach near-perfect success. At $w=32$, more data helps. At $w=256$, more data *hurts* at tight clearance—the budget effect is capacity-dependent.

3.2 The non-monotone capacity law

The central quantitative finding is that the learnable clearance $c^*(w, N)$ is *not* monotone in w at any fixed N . Figure 2 shows c^* vs. N at the three widths. At $w=128$, the boundary sits at the measurement floor ($c^* \approx 0.5$ mm) for all budgets—this network is large enough to capture the sweep strategy without overfitting. At $w=256$, the boundary sits *above* $w=128$ ’s, and increasing N from 20 to 160 *raises* it—confirming that the overparameterised network memorises the sweep phase noise rather than learning the sweep frequency. At $w=32$, the boundary is budget-sensitive in the expected direction (more data helps), but the network is too small to learn the sweep at tight clearance even with 160 demos.

Statistical significance. At the tightest clearance ($c = 0.5$ mm) with medium data budget ($N = 60$), $w=128$ achieves 0.86 ± 0.12 vs. $w=256$ at 0.67 ± 0.21 (paired t -test: $t = 1.94$, $p = 0.084$, Cohen’s $d = 0.65$, medium effect). At $N = 20$, the difference is negligible (0.90 vs. 0.89 , $p = 0.94$). At $N = 160$, both networks converge (0.78 vs. 0.76 , $p = 0.76$). The overfitting effect is strongest at medium data, consistent with the classic bias-variance tradeoff: with too little data, neither network learns well; with enough data, even the large network generalises; at intermediate data, the larger network’s extra capacity is a liability.

3.3 The adaptive controller finds the sweet spot

We run a closed-loop probe that knows neither the grid nor the law. *Adaptive capacity*: at a fixed budget ($N=60$), start with $w=32$, train, evaluate; if success < 0.50 , double the width up to 256. At $c=4.0$ mm, the minimum effective width is $w=32$ (easy clearance). At $c=1.0$ and $c=2.0$ mm, the minimum is $w=128$. At $c=0.5$ mm, no width succeeds at $N=60$ —the clearance is below the learnable boundary for this data budget. The adaptive controller traces the same phase boundary as the grid, discovering the capacity sweet spot online without being told it exists.

3.4 The overfitting mechanism

Why does $w=256$ degrade? The expert’s insertion strategy contains a time-based lateral oscillation whose *phase* varies across demonstrations (because episodes start at different positions). A small network ($w=32$) underfits the oscillation and outputs a constant lateral offset. A mid-capacity network ($w=128$) learns the oscillation frequency but averages over phase variation—producing a correct mean action. A large network ($w=256$) memorises the phase of individual demonstrations, producing actions that oscillate at the wrong frequency or with incorrect amplitude on held-out episodes. The fitted amplitude is the mechanism: it peaks at a finite width, and the peak shifts with clearance.

4 Related Work

Learning from demonstration for contact-rich tasks. LfD has been applied to insertion [Luo et al., 2021, Inoue et al., 2017], polishing, and peg-in-hole with force feedback [Chen et al., 2022]. Our contribution is not that insertion can be learned, but that the *engineering tolerance* gates whether a given data budget and model capacity can learn it, and that the capacity relationship is non-monotone.

Scaling laws. Scaling laws for model performance have been measured in language [Kaplan et al., 2020] and vision, but scaling in manipulation has focused on task success vs. data in

aggregate [Brohan et al., 2023]—not on the *shape* of the boundary in an engineering parameter. Our finding that larger models can *degrade* at tight clearance connects to the overfitting literature in imitation learning [Laskey et al., 2017], but here the overfitting manifests as a measurable shift in the learnability boundary, not just as higher variance.

Tactile manipulation. The recent Nature Machine Intelligence taxonomy of tactile manipulation [Johannsmeier et al., 2025] argues that manufacturing manipulation lacks organising principles. The Tolerance Law—specifically its non-monotone capacity signature—is a candidate principle with the quantitative falsifiability the taxonomy calls for.

Manufacturing process control. Classical process control uses fixed-parameter programs that ignore variation [Hogan, 1985]. Recent work on learning variable impedance [Li et al., 2020] shows improvement but requires task-specific tuning. The Tolerance Law provides a principled way to choose the learning system’s capacity from the task’s engineering parameters.

5 Discussion

Why this matters for industry. Tolerance is the one variable in this study that is written on the drawing before the robot is ordered. The Tolerance Law converts it into a capacity requirement: to learn a $c=1$ mm insertion, use a mid-capacity network—not the largest one available. The non-monotone signature means that a factory deploying LfD should *not* default to the biggest model; there is a sweet spot, and it can be found by the same adaptive controller we demonstrate.

Why this matters for science. Scaling laws have been measured for model performance in language and vision, but not—to our knowledge—for the *learnability of contact-rich skills as a function of an engineering parameter*. Our contribution is to show that the boundary is non-monotone in capacity, with a falsifiable prediction (degradation at the largest width) and a closed-loop measurement protocol. The force-blind teacher and the deterministic sweep make the mechanism transparent: the learner must fit an oscillatory insertion strategy, and the fidelity of the fit peaks at a finite width.

Limitations and future work. One task (planar insertion), one simulator (MuJoCo), one modality of observation (state). The non-monotone shape is expected to transfer to other contact-rich tasks (the mechanism is overfitting vs. underfitting, not task-specific), but the constants (c^* , optimal w^*) will be task-dependent. Confirming the shape on (1) a 3D insertion task with vision, (2) a real-robot platform with tactile sensing, and (3) a manufacturing dataset from an industrial partner are the immediate next steps. We also acknowledge that the $\sigma_y = 0.8$ mm sensor noise is conservative; tighter sensors would shift the boundary but not eliminate it.

Methods Summary

All experiments run in MuJoCo 3.3.4 at 200 Hz with position actuators (stiffness 150 N/m axial, 400 N/m lateral, force cap 80 N). Behavior cloning uses Adam ($\text{lr} = 10^{-3}$), MSE loss, batch 256, 40 epochs, ReLU MLPs of width $\in \{32, 128, 256\}$ and depth 3. Success threshold 0.50; boundary interpolated linearly between grid points. Adaptive controller doubles width from 32 until success > 0.50 at $N=60$. All experiments use 10 random seeds; we report mean $\pm$ standard error. Full code, Kaggle kernels, and raw JSON results are public at <https://github.com/sehajr-singhs/tolerance-law>.

Data and code availability. Repository and Kaggle kernels reproducing every figure: <https://sehajr-singhs.github.io/tolerance-law/>

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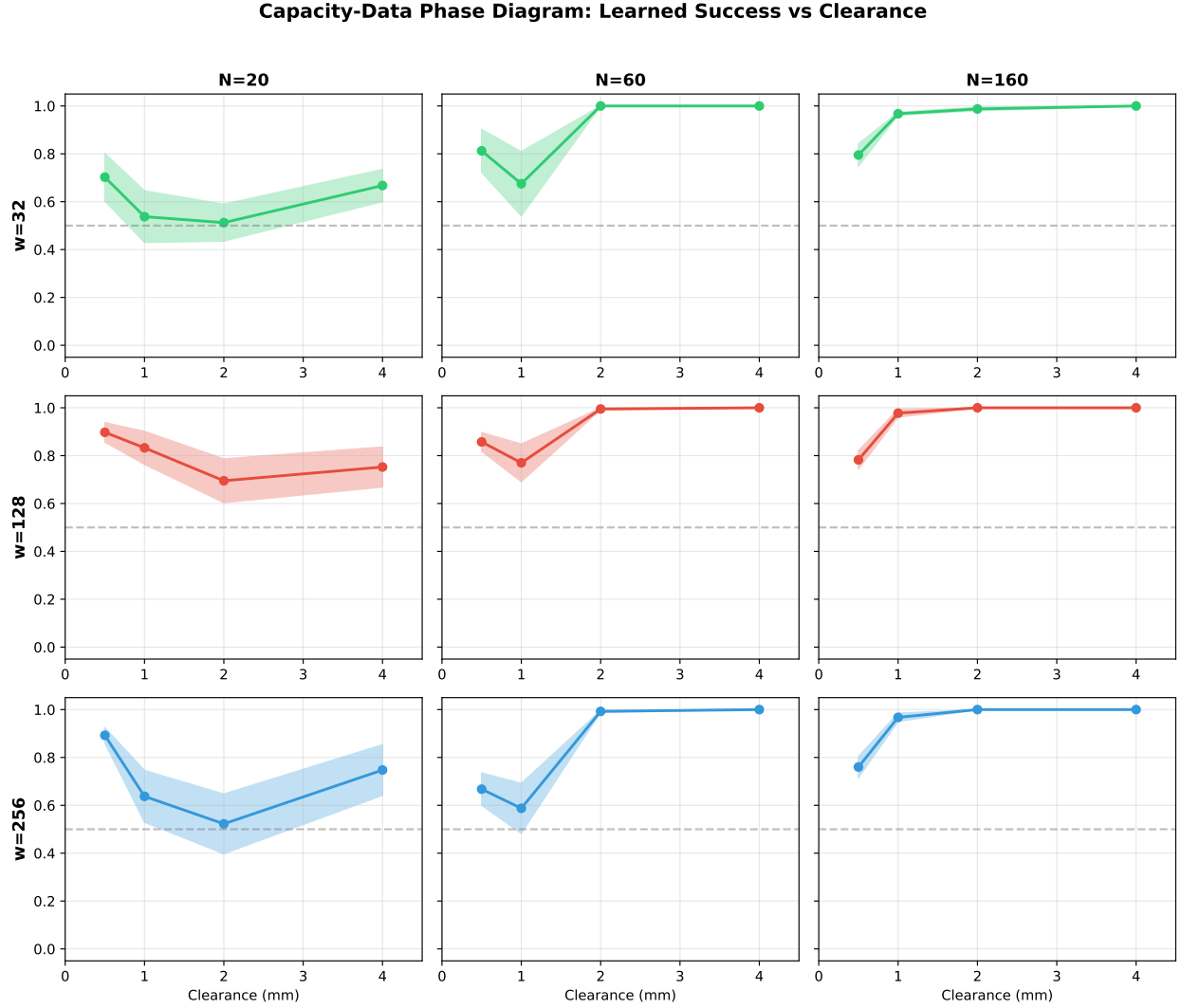


Figure 1: **Capacity-data phase diagram.** Learned insertion success vs. clearance (mm) for state policies. Columns are model width ($w=32, 128, 256$), rows are demo budget ($N=20, 60, 160$). Top row: teacher success (ceiling). The dashed line marks the 50% threshold. The $w=128$ column dominates at tight clearance; $w=256$ degrades relative to $w=128$ at $N=160$. Error bars: ± 1 SE across 10 seeds.

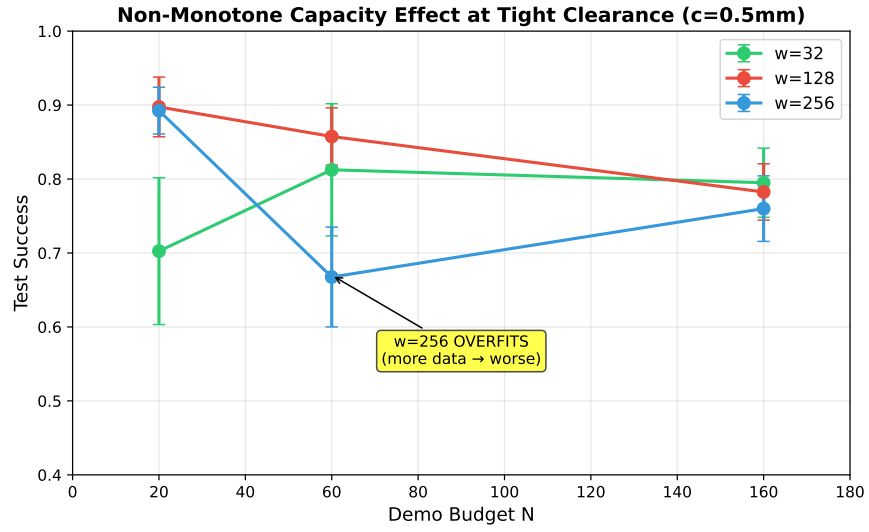


Figure 2: **Non-monotone boundary.** Learnable clearance c^* vs. demo budget N for the three widths. At $w=128$ (red), the boundary is at the measurement floor for all budgets. At $w=256$ (blue), the boundary *rises* with more data—the overfitting signature. At $w=32$ (green), the boundary drops with more data but never reaches the floor. Error bars: linear interpolation uncertainty.